DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503

1. TITLE	2. IDENTIFICATION NUMBER
Airframe Rigidity Test Report	DI-MISC-81387

3. DESCRIPTION/PURPOSE

3.1 This report provides the data for detailed evaluation of the air vehicle modal vibration characteristics and rigidity properties and comparison with calculated results. The principal use is to validate or update analytical models.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
940211	F/ASC-ENFSL		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) contains the format and content preparation instructions for the data product generated by the specific and discrete work task requirements as delineated in the contract.

7.2 This DID is applicable in any contract where a ground vibration test or other airframe rigidity test is required.

(Continued on page 2)

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F6999

10. PREPARATION INSTRUCTIONS

10.1 Format. Contractor's format is acceptable.

10.2 Content.

10.2.1 Air vehicle ground vibration test. This section shall contain the following:

a. Descriptions and photographs of the air vehicle suspension system, the instrumentation, and the air vehicle configurations.

b. Description of test procedures, data acquisition, and methods of data reduction/analysis.

c. Shaker locations.

d. Mode shape measurement locations.

(Continued on page 2)

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Block 7. Application/Interrelationship (Continued)

7.3 This DID supersedes DI-T-30734.

Block 10. Preparation Instructions (Continued)

e. Plots of the transfer function versus frequency for strategically located vibration pickups.

f. The natural frequency, mode shapes, structural damping, and node line locations for all obtained structural modes of vibration that may be important with respect to flutter and other dynamic analyses.

g. A comparison of the measured air vehicle modal data (frequencies, mode shapes, and node lines) with those as predicted by the analytical model.

10.2.2 Aeroservoelastic ground test. When applicable, this section shall contain the following:

a. Test description.

b. Comparison of the test results with corresponding analysis.

10.2.3 Component test. When applicable, this section shall contain results of the following tests:

a. Component ground vibration tests on control surfaces, pylons, flaps, horizontal stabilizers, canards, control surface actuators, control surface dampers, racks for supporting stores, weapons bay doors, landing gear doors, and landing gear.

b. Component stiffness tests.

c. Control surface, tab, and actuator rigidity, free play, and wear tests.

d. Balance weight attachment verification tests.

e. Mass property measurements of control surfaces, tabs, and external stores.

f. Flutter damper qualification and wear tests.

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